Vivekbhai Radadiya

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Experience

Data Scientist — June 2025 – Present
Community Dreams Foundation — *Remote, United States*

- Developed an NLP pipeline using SpaCy and transformers to extract red flags from 190K+ municipal bond disclosures, reducing analyst review time by 38%, improving early detection of structural risk patterns in fragmented documents.
- Built anomaly detection models using autoencoders and Isolation Forests across municipal portfolios, which increased risk flagging accuracy by 32% and enabled analysts to surface subtle signals missed in traditional rule-based reviews.
- Deployed Streamlit and PowerBI dashboards visualizing credit metrics from bond data, reducing manual report generation time by 44%, allowing research teams to prioritize high-risk issuers across 2 million securities in real time.
- Engineered an AWS Lambda and S3-based ETL pipeline processing 2.1 million bond records daily, reducing batch latency by 56% and enabling continuous updates for live credit scoring dashboards.
- Trained LSTM models using economic and issuer-level signals to forecast credit deterioration, improving predictive precision by 23% compared to logistic regression models used in baseline credit surveillance tasks.
- Fine-tuned transformer-based models to identify financial disclosure anomalies, increasing extraction accuracy by 35% and enabling automated detection of fiscal irregularities from unstructured municipal offering documents.

AI/ML Research Assistant – Data Science Team — March 2024 – May 2025
Northeastern University — *Boston, United States*

- Applied BERT embeddings and sequence tagging on 11,000+ budget disclosures to flag credit deterioration, increasing lead time on high-risk identification by 31% compared to manual screening approaches.
- Devised a bond ranking recommendation engine using matrix factorization on credit resilience signals, boosting Sharpe ratio by 12% in portfolio simulations run on backtested municipal market data from 2015–2023.
- Performed anomaly detection with PCA and time-series clustering on fragmented budget reports, revealing hidden risks in 14% of issuers and informing internal downgrade watchlists across long-tail municipalities.
- Created Flask APIs for internal access to risk metrics and bond forecasts, reducing R&D feedback time by 29% and supporting more responsive model validation workflows among research analysts.

Senior Data Scientist — May 2021 – June 2023
Niqox Technologies Pvt. Ltd. — *Surat, India*

- Designed a sophisticated risk scoring framework that incorporated fiscal indicators from 50+ cities; this system has become the benchmark for assessing credit distress, leading to enhanced portfolio restructuring strategies for clients.
- Optimized Snowflake SQL workflows for 5.2TB of fixed income data, reducing average query time from 9.8s to 1.2s, which accelerated credit scoring computations and supported real-time model updates.
- Extracted financial event signals from 3.2 million regional news articles using NLP, improving recall of fiscal stress indicators by 22% and expanding credit surveillance beyond formal disclosures.

Data Scientist — January 2020 – April 2021
Niqox Technologies Pvt. Ltd. — *Surat, India*

- Applied OCR and regex techniques to extract covenants from 120,000+ municipal bond PDFs, automating 81% of document review and enabling analysts to flag compliance violations with minimal manual effort.
- Formulated PyTorch models for municipal bond credit scoring, increasing multi-class classification accuracy by 24% and outperforming tree-based models in identifying low-grade securities in historical datasets.
- Automated model retraining on Azure ML by orchestrating pipelines to ingest updated macroeconomic data, reducing update cycles by 35% and enabling models to remain current without manual intervention.

Education

Master of Science in Data Analytics Engineering — May 2025
Northeastern University – College of Engineering — *Boston, United States*

Coursework: Cloud Computing, Data Warehousing, MLOps, Machine Learning, Neural Networks and Deep Learning

Bachelor of Science in Mathematics — September 2020
Veer Narmad South Gujarat University - Department of Mathematics — *Surat, India*

Technical Skills

Languages: Python (object-oriented programming), SQL, Java, Scala, R
Tools: AWS, Google Cloud, Azure, Tableau, PowerBI, Streamlit, Flask, Git, Azure DevOps, Snowflake, BigQuery, NLTK
Frameworks/Libraries: scikit-learn, TensorFlow, PyTorch, Seaborn, Matplotlib, SpaCy, Hugging Face Transformers
Certifications: AWS Certified Solutions Architect – Associate, Microsoft Certified: Azure Data Scientist Associate